UNIT-III
Data Converters

Introduction

- The signal can be analog or digital by nature.
- Sometimes there is a need to convert analog signal into digital equivalent and vice versa.
- The electronics circuits used for such a conversion are called as **data converters**.
- Most of the physical quantities, such as temperature, pressure, displacement, vibrations, etc., are available in analog form.
- These quantities are represented accurately in analog form, but it is difficult to process, store or transmit the analog signal because error gets introduced easily due to noise.

- The digital representation of a signal makes it possible to process simpler and transmit more easily.

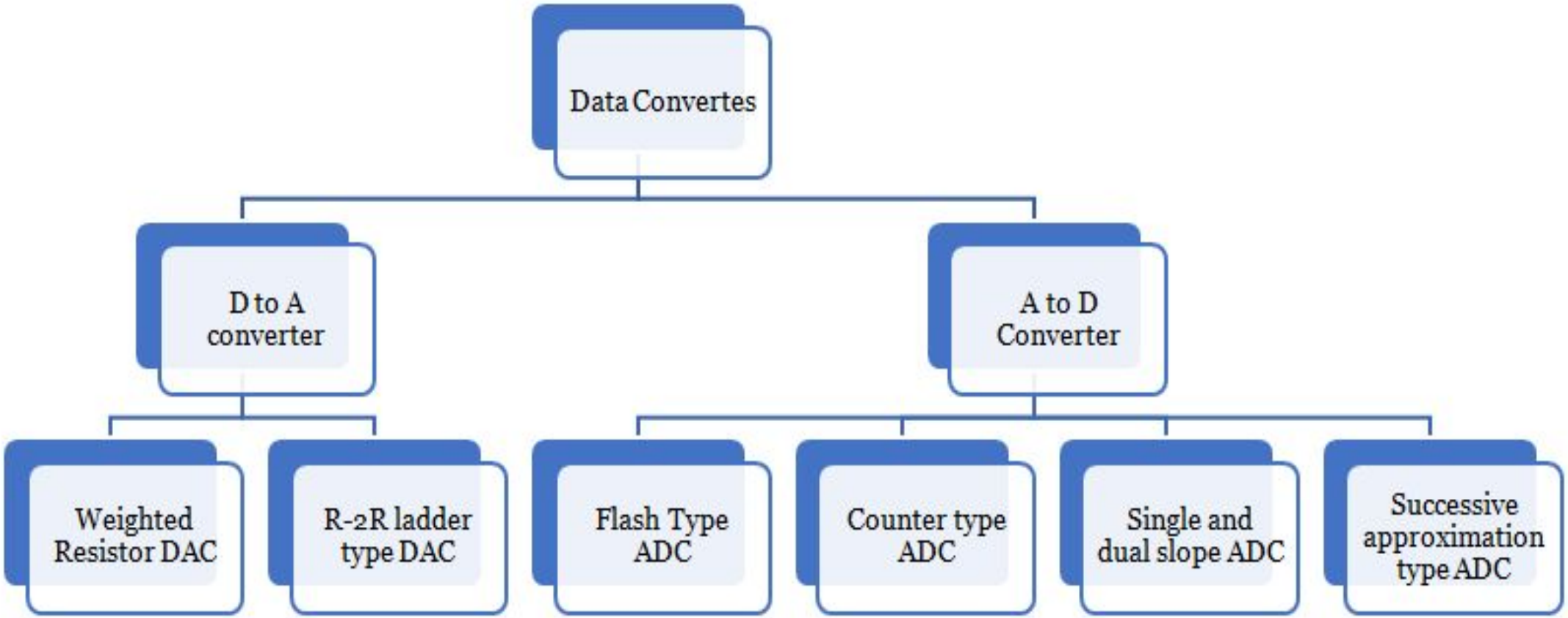
- Therefore A to D conversion is necessary. now once the processing, transmission, etc. is done, the signal should be back to its analog form, for which the D to A conversion is essential.

- Both A to D and D to A circuits are called data converters, and they are available in IC form.

Types of Data Converters

- The data converters are basically of two types:

- Digital to Analog Converters (DACs)
- Analog to Digital Converters (ADCs)



Classification of Data Converters

Digital to Analog Converters

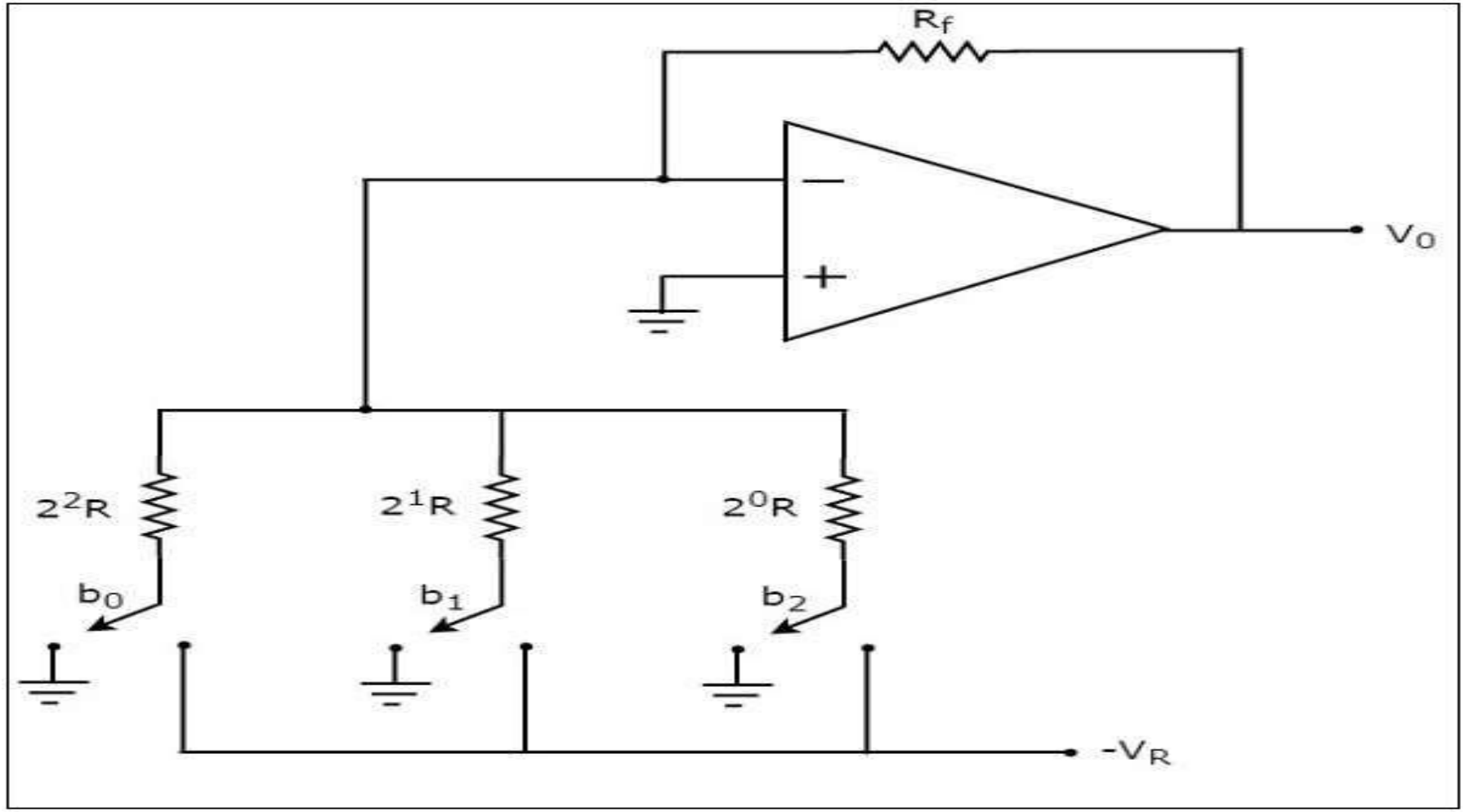
- A **Digital to Analog Converter (DAC)** converts a digital input signal into an analog output signal. The digital signal is represented with a binary code, which is a combination of bits 0 and 1.
- A Digital to Analog Converter (DAC) consists of a number of binary inputs and a single output. In general, the **number of binary inputs** of a DAC will be a power of two.

The block diagram of DAC



Weighted Resistor DAC

- A weighted resistor DAC produces an analog output, which is almost equal to the digital (binary) input by using **binary weighted resistors** in the inverting adder circuit.



- Let the 3-bit binary input be b2b1b0. Here, the bits b2 and b0 denote the Most Significant Bit (MSB) and Least Significant Bit (LSB), respectively.
- The digital switches will be connected to ground when the corresponding input bits are equal to 0. Similarly, the digital switches shown in the above figure will be connected to the negative reference voltage, −VR, when the corresponding input bits are equal to 1.
- The non-inverting input terminal of an op-amp is connected to ground. That means zero volts is applied at the non-inverting input terminal of the op-amp.
- According to the **virtual short concept**, the voltage at the inverting input terminal of the opamp is same as that of the voltage present at its non-inverting input terminal. So, the voltage at the inverting input terminals node will be zero volts.

The **nodal equation** at the inverting input terminals node is:

$$\frac{0+V_R b_2}{2^0 R}+\frac{0+V_R b_1}{2^1 R}+\frac{0+V_R b_0}{2^2 R}+\frac{0-V_0}{R_f}=0$$

$$=>\frac{V_0}{R_f}=\frac{V_R b_2}{2^0 R}+\frac{V_R b_1}{2^1 R}+\frac{V_R b_0}{2^2 R}$$

$$=> V_0=\frac{V_R R_f}{R}\left\{\frac{b_2}{2^0}+\frac{b_1}{2^1}+\frac{b_0}{2^2}\right\}$$

Substituting, $R=2R_f$ in above equation.

$$=> V_0=\frac{V_R R_f}{2R_f}\left\{\frac{b_2}{2^0}+\frac{b_1}{2^1}+\frac{b_0}{2^2}\right\}$$

$$=> V_0=\frac{V_R}{2}\left\{\frac{b_2}{2^0}+\frac{b_1}{2^1}+\frac{b_0}{2^2}\right\}$$

- The above equation represents the output voltage equation of a 3-bit binary weighted resistor DAC. Since the number of bits is three in the binary (digital) input, we will get seven possible values of output voltage by varying the binary input from 000 to 111 for a fixed reference voltage, VR.
- The generalized output voltage equation of an N-bit binary weighted resistor DAC, based on the output voltage equation of a 3-bit binary weighted resistor DAC as

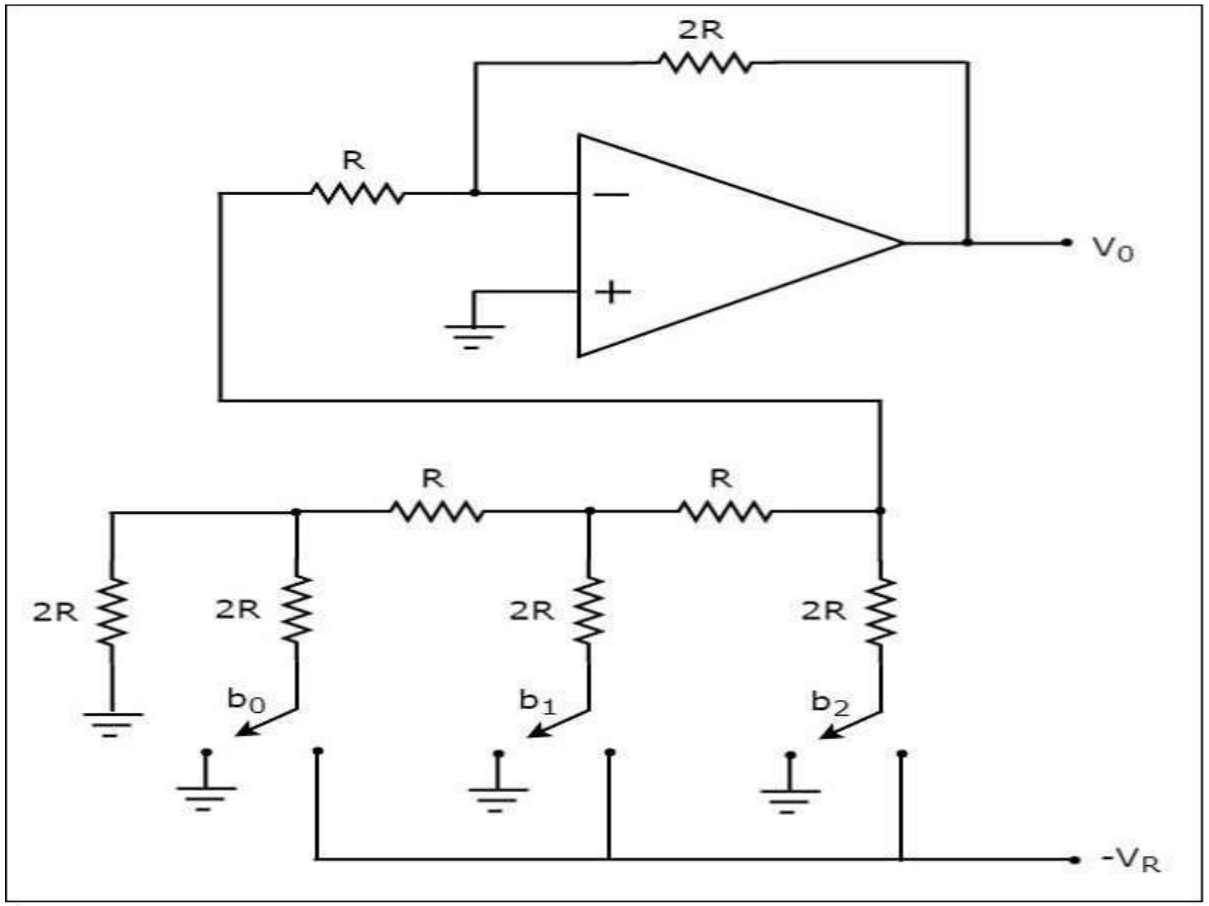
$$=> V_0 = \frac{V_R}{2}\left\{\frac{b_{N-1}}{2^0} + \frac{b_{N-2}}{2^1} + \ldots + \frac{b_0}{2^{N-1}}\right\}$$

The disadvantages of a binary weighted resistor DAC are as follows –

- The difference between the resistance values corresponding to LSB & MSB will increase as the number of bits present in the digital input increases.
- It is difficult to design more accurate resistors as the number of bits present in the digital input increases.

R-2R Ladder DAC

- The R-2R Ladder DAC overcomes the disadvantages of a binary weighted resistor DAC.
- The R-2R Ladder DAC produces an analog output, which is almost equal to the digital (binary) input by using a **R-2R ladder network** in the inverting adder circuit.



- Let the 3-bit binary input be b2b1b0. Here, the bits b2 and b0 denote the Most Significant Bit (MSB) and Least Significant Bit (LSB), respectively.
- The digital switches shown in the above figure will be connected to ground when the corresponding input bits are equal to 0.
- Similarly, the digital switches will be connected to the negative reference voltage, −VR, when the corresponding input bits are equal to 1.
- It is difficult to get the generalized output voltage equation of an R-2R Ladder DAC.

The **advantages** of an R-2R Ladder DAC are as follows −

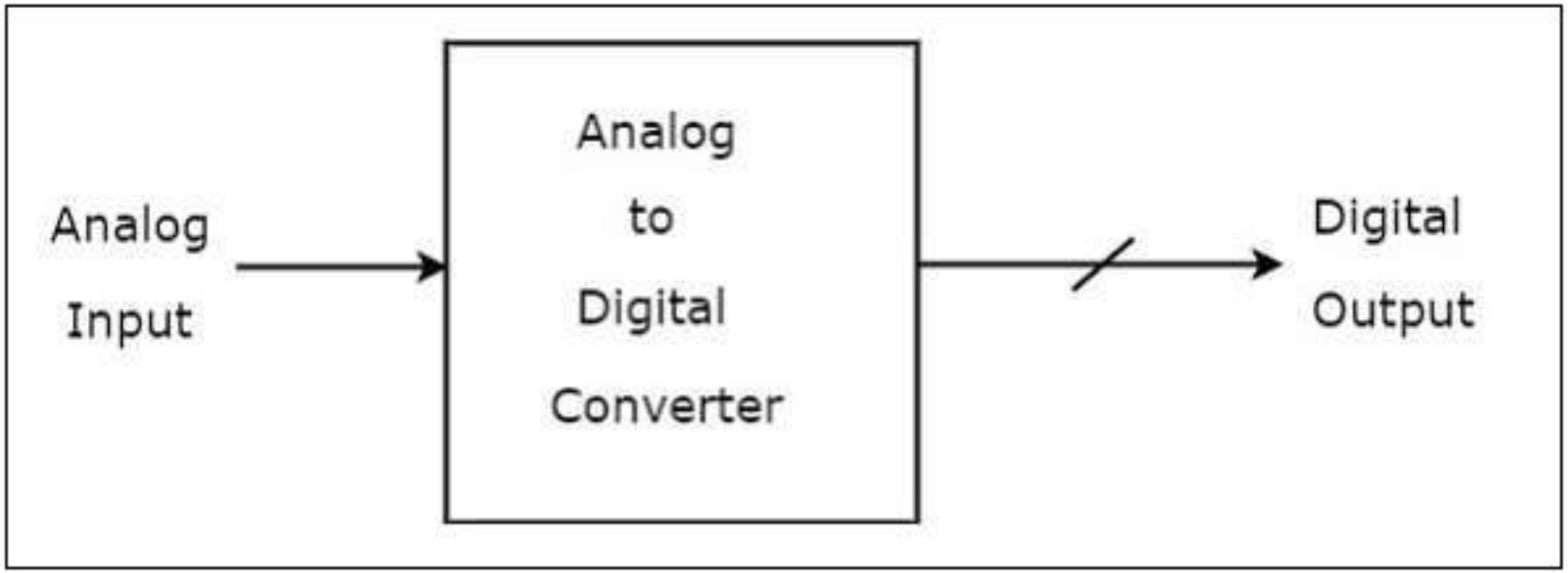
- R-2R Ladder DAC contains only two values of resistors: R and 2R. So, it is easy to select and design more accurate resistors.
- If more number of bits are present in the digital input, then we have to include the required number of R-2R sections additionally.

Analog to Digital Converter

- An **analog-to-digital converter**, also known as **ADC**, is a digital circuit used to convert analog signals into digital format.
- The conversion of analog signals into digital format is crucial for their processing with the help of digital systems like microprocessors, microcontrollers, digital signal processors (DSPs), etc.
- Therefore, ADCs are important components in many digital systems, such as computers and other digital devices.
- An Analog to Digital Converter **(ADC)** converts an analog signal into a digital signal. The digital signal is represented with a binary code, which is a combination of bits 0 and 1.

What is an Analog to Digital Converter?

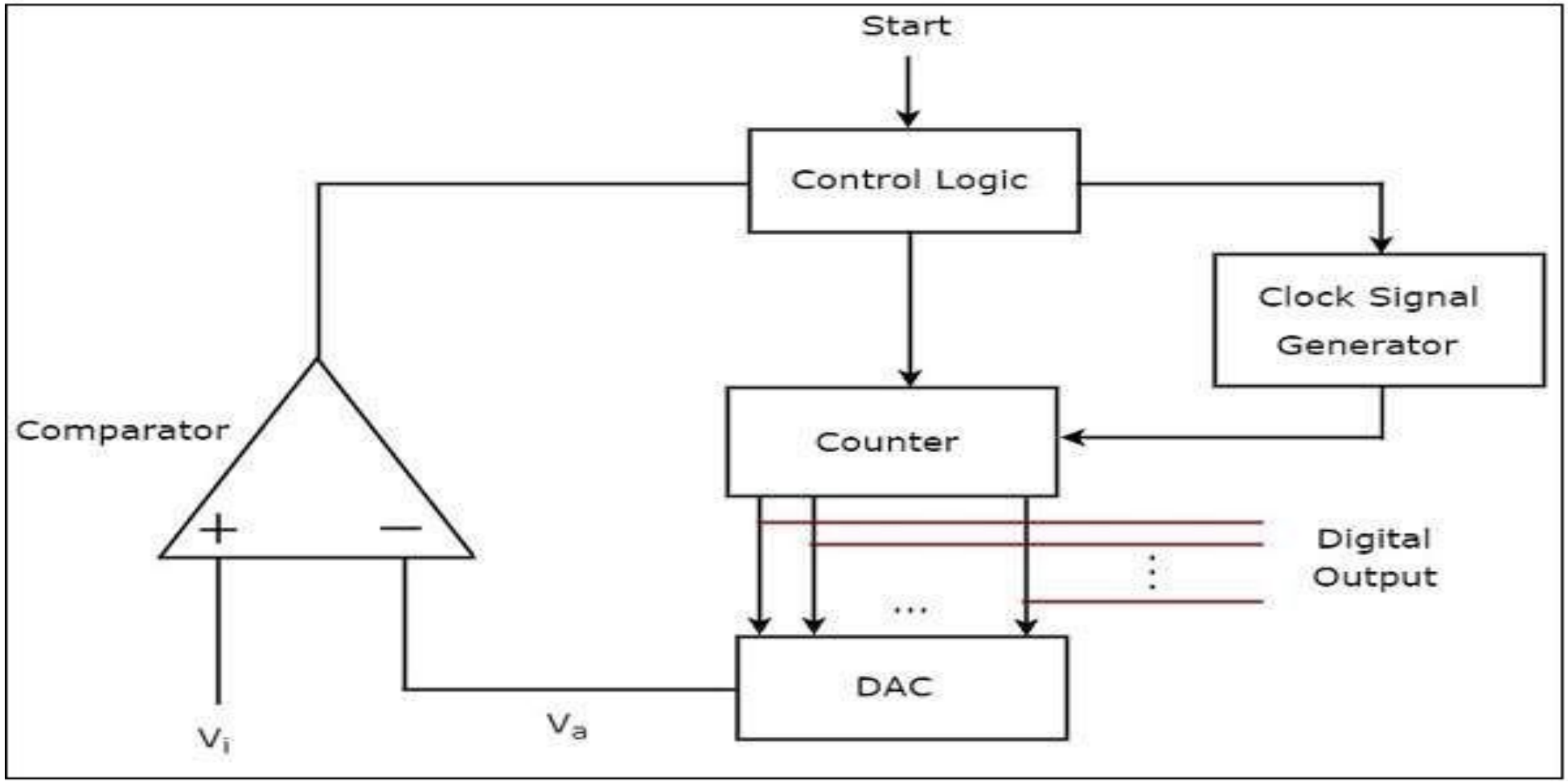
- An analog to digital converter is a digital circuit designed to perform the conversion of analog signals into digital data format. It is also known as ADC. Analog-to-digital converters are essential components in digital systems such as computers, data processors, and digital communication systems.



- An Analog to Digital Converter (ADC) consists of a single analog input and many binary outputs. In general, the number of binary outputs of an ADC will be a power of two.
- There are two types of ADCs: Direct type ADCs and Indirect-type ADCs.
- If the ADC performs the analog to digital conversion directly by utilizing the internally generated equivalent digital (binary) code for comparing with the analog input, then it is called as **Direct type ADC**.
- The following are **examples** of Direct type ADCs –
- Counter type ADC
- Successive Approximation ADC
- Flash type ADC

Counter type ADC

- A **counter type ADC** produces a digital output, which is approximately equal to the analog input by using counter operation internally.



- The counter type ADC mainly consists of 5 blocks: Clock signal generator, Counter, DAC, Comparator, and Control logic.

The working of a counter type ADC is as follows –

- The control logic resets the counter and enables the clock signal generator to send the clock pulses to the counter when it receives the start commanding signal.
- The counter gets incremented by one for every clock pulse, and its value will be in binary (digital) format. The output of the counter is applied as an input of DAC.
- DAC converts the received binary (digital) input, which is the output of the counter, into an analog output. Comparator compares this analog value, Va, with the external analog input value Vi

- The output of the comparator will be 1 as long as it is greater than. The operations mentioned in the above two steps will be continued as long as the control logic receives 1 from the output of the comparator.

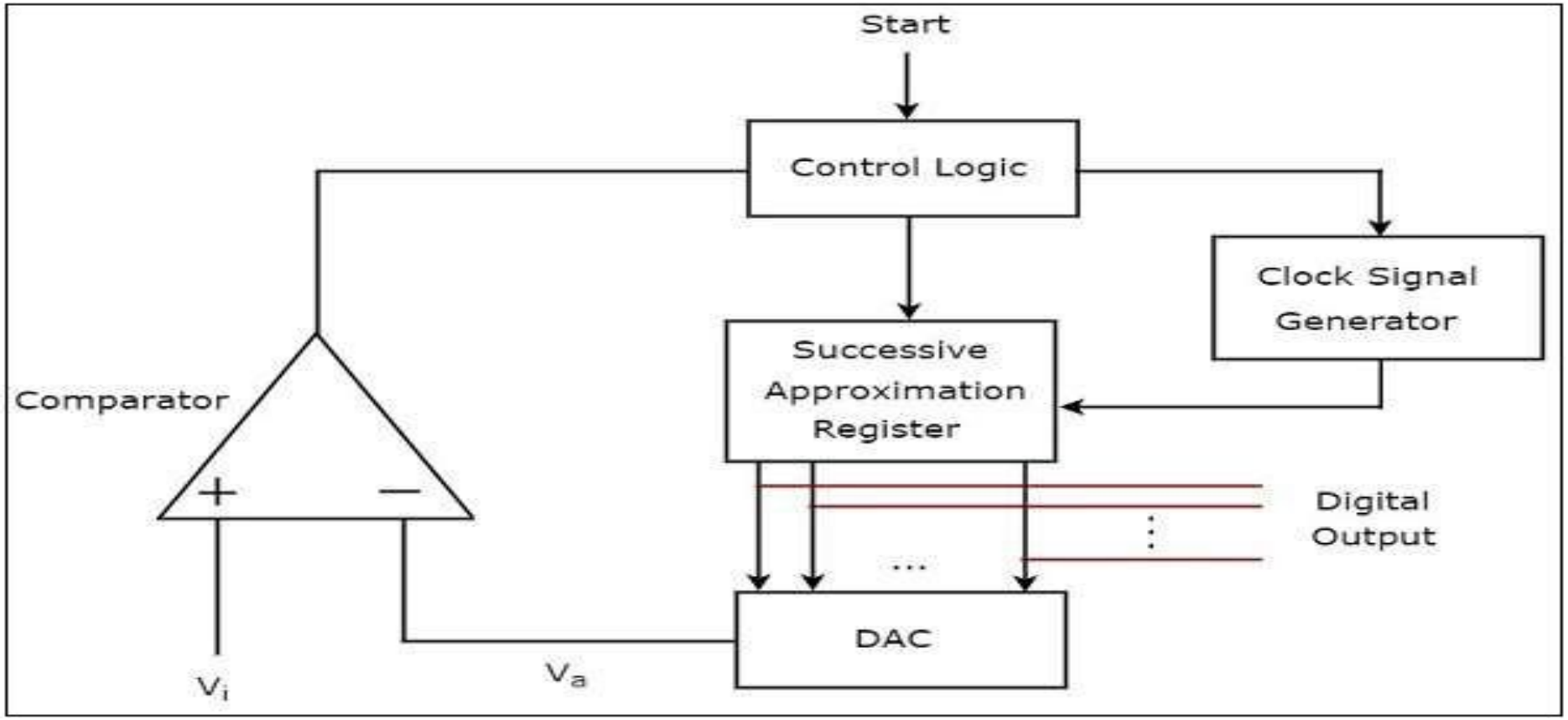
- The output of the comparator will be 0 when Vi is less than or equal to Va. So, the control logic receives 0 from the output of the comparator.

- Then, the control logic disables the clock signal generator so that it doesn’t send any clock pulse to the counter.

- At this instant, the output of the counter will be displayed as the digital output. It is almost equivalent to the corresponding external analog input value Vi.

Successive Approximation ADC

- A **successive approximation type ADC** produces a digital output, which is approximately equal to the analog input by using the successive approximation technique internally.



- The successive approximation ADC mainly consists of 5 blocks– Clock signal generator, Successive Approximation Register (SAR), DAC, comparator, and Control logic.

The working of a successive approximation ADC is as follows –

- The control logic resets all the bits of SAR and enables the clock signal generator to send the clock pulses to SAR when it receives the start commanding signal.

- The binary (digital) data present in SAR will be updated for every clock pulse based on the output of the comparator. The output of SAR is applied as an input to DAC.

- DAC converts the received digital input, which is the output of SAR, into an analog output. The comparator compares this analog value, Va, with the external analog input value Vi.

- The output of a comparator will be 1 as long as Vi is greater than Va. Similarly, the output of the comparator will be 0 when Vi
- is less than or equal to Va.

- The operations mentioned in the above steps will be continued until the digital output is a valid one.

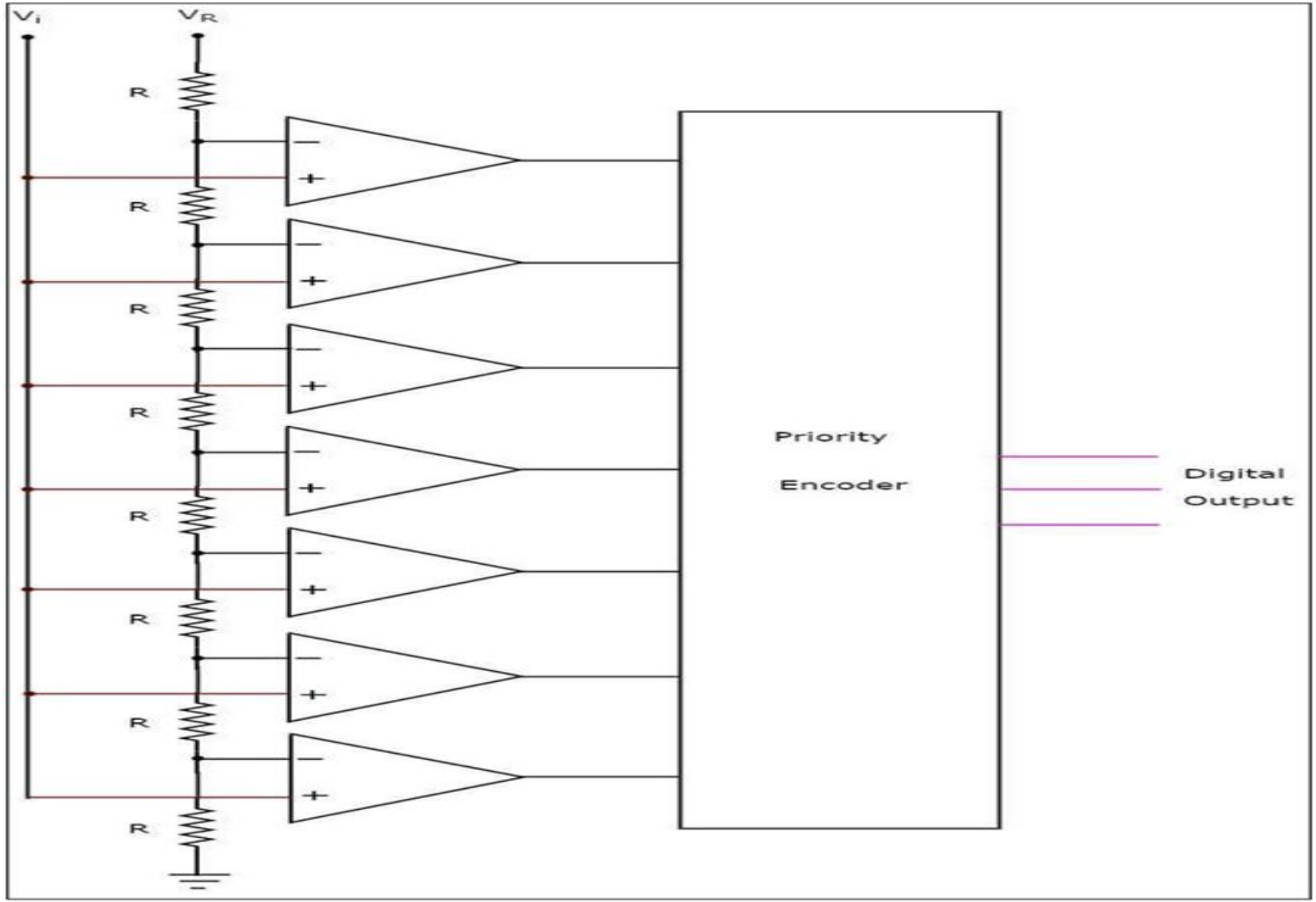
- The digital output will be a valid one, when it is almost equivalent to the corresponding external analog input value Vi

Flash type ADC

- A **flash type ADC** produces an equivalent digital output for a corresponding analog input in no time. Hence, flash-type ADC is the fastest ADC.
- The 3-bit flash type ADC consists of a voltage divider network, 7 comparators, and a priority encoder.

The working of a 3-bit flash type ADC is as follows.

- The voltage divider network contains 8 equal resistors. A reference voltage VR is applied across that entire network with respect to the ground.
- The voltage drop across each resistor from bottom to top with respect to ground will be the integer multiples (from 1 to 8) of VR8.
- The external input voltage Vi is applied to the non-inverting terminal of all comparators. The voltage drop across each resistor from bottom to top with respect to ground is applied to the inverting terminal of comparators from bottom to top.


Vi
VR
R
R
R
R
R
R
R
R
Priority
Encoder
Digital
Output

- At a time, all the comparators compare the external input voltage with the voltage drops present at the respective other input terminal. That means the comparison operations take place in each comparator.

- The output of the comparator will be 1 as long as Vi is greater than the voltage drop present at the respective other input terminal. Similarly, the output of the comparator will be 0 when Vi is less than or equal to the voltage drop present at the respective other input terminal.

- All the outputs of comparators are connected as the inputs of the priority encoder. This priority encoder produces a binary code (digital output), which corresponds to the high-priority input that has 1.

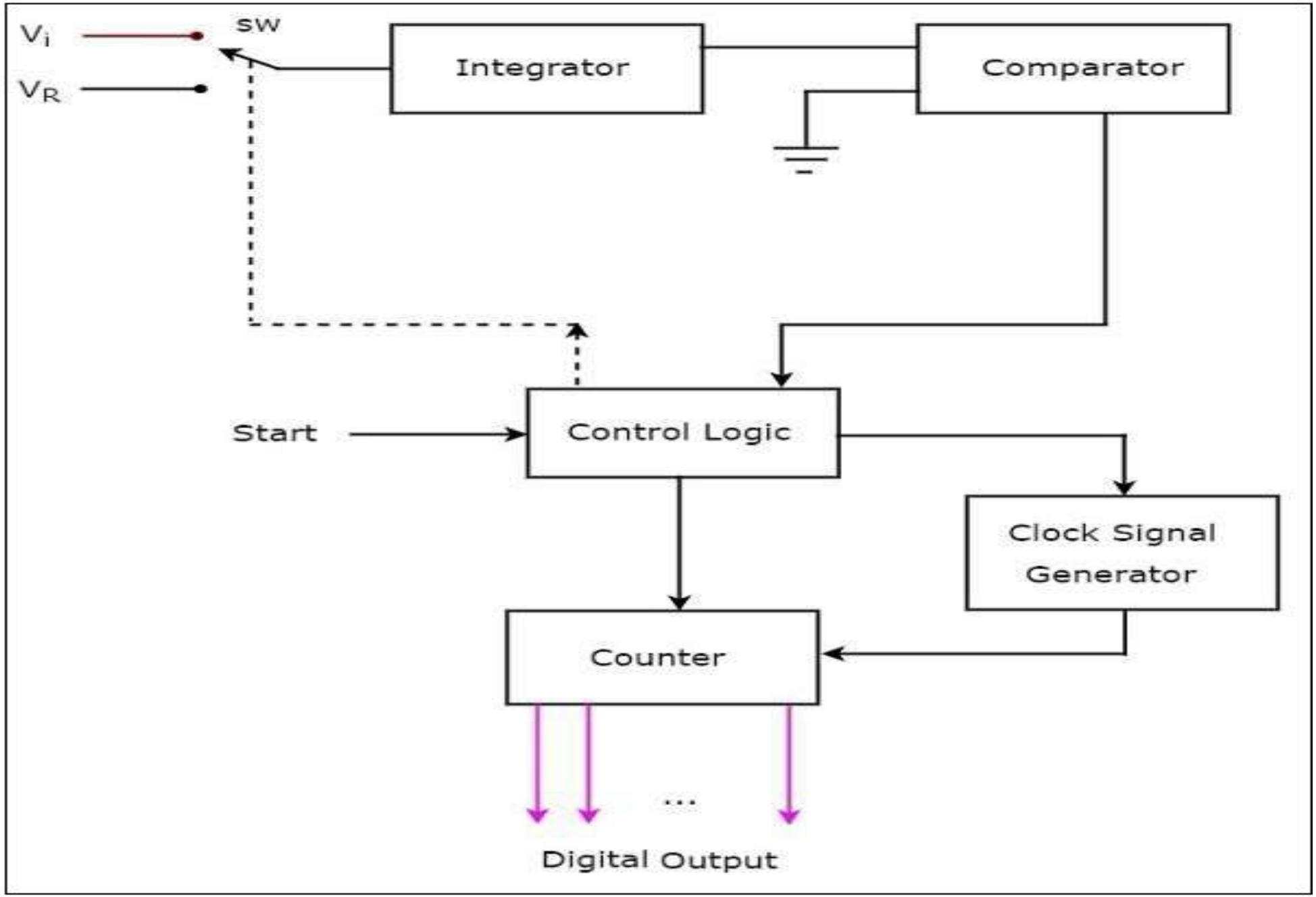
- Therefore, the output of the priority encoder is nothing but the binary equivalent (digital output) of the external analog input voltage, Vi.
- The flash type ADC is used in applications where the conversion speed of analog input into digital data should be very high.

Indirect Type ADC

- If an ADC performs the analog to digital conversion by an indirect method, then it is called an **Indirect type ADC**.
- In general, first it converts the analog input into a linear function of time (or frequency), and then it produces the digital (binary) output.
- The dual slope ADC is the best **example** of an Indirect type ADC.

Dual Slope ADC

- As the name suggests, a **dual slope ADC** produces an equivalent digital output for a corresponding analog input by using two (dual) slope technique.


V_i
V_R
SW
Integrator
Comparator
Start
Control Logic
Clock Signal Generator
Counter
...
Digital Output

- The dual slope ADC mainly consists of 5 blocks: Integrator, Comparator, Clock signal generator, Control logic, and Counter.

The working of a dual slope ADC is as follows –

- The control logic resets the counter and enables the clock signal generator to send the clock pulses to the counter when the start commanding signal.

- Control logic pushes the switch sw to connect to the external analog input voltage Vi when the start commanding signal. This input voltage is applied to an integrator.

- The output of the integrator is connected to one of the two inputs of the comparator, and the other input of the comparator is connected to ground.

- Comparator compares the output of the integrator with zero volts (ground) and produces an output, which is applied to the control logic.

- The counter gets incremented by one for every clock pulse, and its value will be in binary (digital) format. It produces an overflow signal to the control logic when it is incremented after reaching the maximum count value. At this instant, all the bits of the counter will have zeros only.

- Now, the control logic pushes the switch sw to connect to the negative reference voltage –Vref. This negative reference voltage is applied to an integrator. It removes the charge stored in the capacitor until it becomes zero.

- At this instant, both the inputs of a comparator are at zero volts. So, the comparator sends a signal to the control logic.
- Now, the control logic disables the clock signal generator and retains (holds) the counter value.
- The counter value is proportional to the external analog input voltage.
- At this instant, the output of the counter will be displayed as the digital output.
- It is almost equivalent to the corresponding external analog input value Vi.
- The dual slope ADC is used in applications where **accuracy** is more important while converting analog input into its equivalent digital (binary) data.

Specifications

Resolution

- Resolution is the minimum amount of change needed in an analog input voltage for it to be represented in a binary (digital) output. It depends on the number of bits that are used in the digital output.
- **Resolution** can also be defined as the ratio of the maximum analog input voltage that can be represented in binary and the equivalent binary number. Mathematically, resolution can be represented as

$$Resolution = \frac{V_{FS}}{2^N - 1}$$

VFS is the full-scale input voltage or the maximum analog input voltage,

N is the number of bits that are present in the digital output.

Conversion Time: The amount of time required for a data converter to convert the data (information) of one form into its equivalent data in another form is called **conversion time**.

- **Settling Time:** Time taken for the output to reach a stable value within a specified percentage.
- **Monotonicity:** Ensures the output voltage always increases or stays constant as the digital input increases.
- **Accuracy:** Absolute accuracy is the maximum deviation between the actual converter output and the ideal converter output. Relative accuracy is the maximum deviation after gain and offset errors have been removed.